

PRESENTATION

By Wajiha Fareed

MEDIGUIDE AI

- MediGuide AI is an AI-based medical symptom assessment system.
- It collects basic patient information and symptoms.
- It uses Python, LangChain, OpenAI LLM, and Streamlit.
- It provides structured, safety-focused preliminary guidance.
- It is an educational prototype, not a replacement for a doctor or professional diagnosis

PROBLEM STATEMENT

- People often search their **symptoms** on random websites.
- Online medical information may be **unreliable** or **confusing**.
- Different sources can provide **inconsistent information**.
- This can create **anxiety** and **uncertainty** for users.
- MediGuide AI provides **structured, understandable, and safety-focused** guidance

OBJECTIVES

- **Collect** basic patient information and symptoms.
- **Analyse** the provided symptoms using an AI model.
- **Generate** clear and structured medical guidance.
- **Identify** the appropriate urgency level.
- **Recommend** suitable next steps and highlight important warning signs.

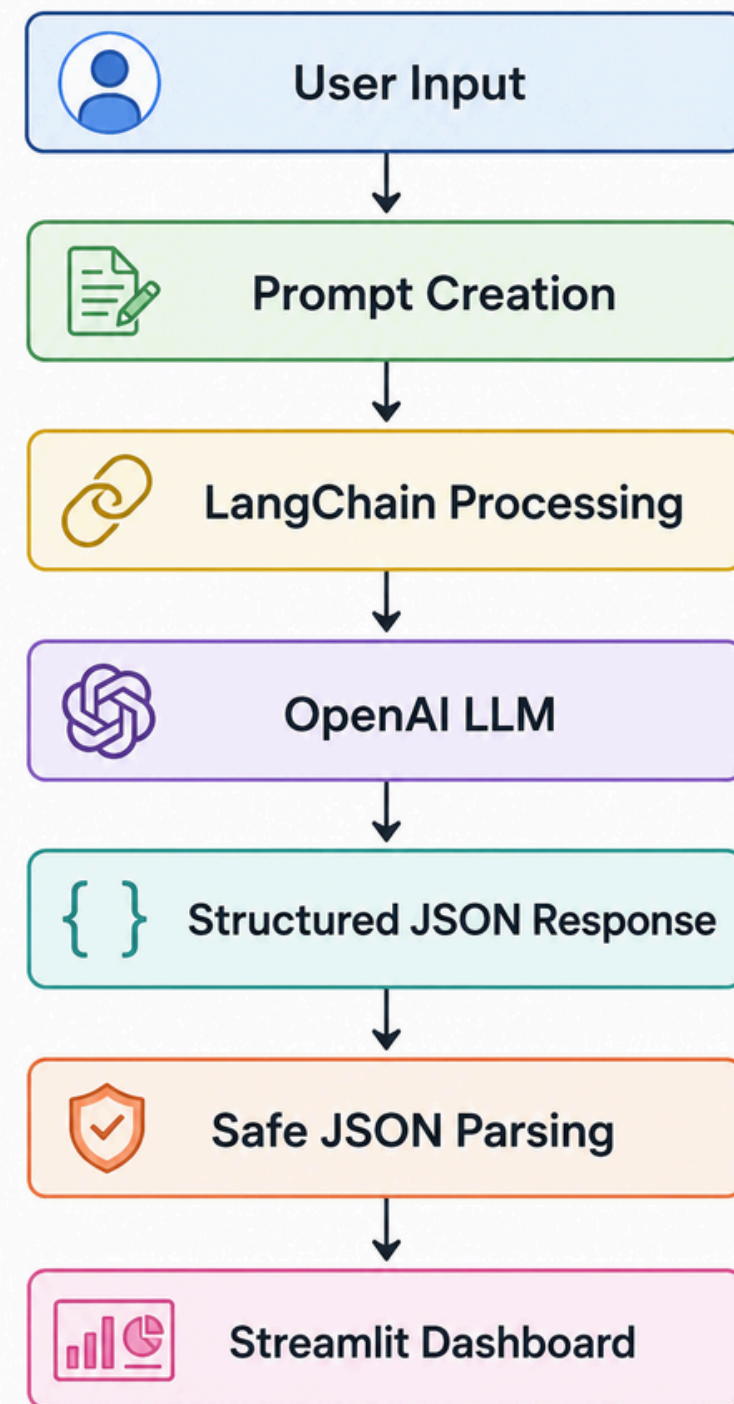
TECHNOLOGIES USED

- **Python** — Main programming language for the application.
- **LangChain** — Used to build and manage the LLM workflow.
- **OpenAI LLM** — Analyses the provided patient information and symptoms.
- **Streamlit** — Creates the interactive user interface and results dashboard.
- **InMemoryCache & SQLiteCache** — Improve performance by caching repeated requests.

USER INPUTS

- Age
- Gender
- Symptoms
- Duration of symptoms
- Severity level (1–10)
- Preferred language
- Existing medical conditions
- Current medications
- Additional notes

SYSTEM WORKFLOW



RESULTS DASHBOARD

- Symptom Summary
- AI-Generated Health Information
- Possible Conditions
- Urgency Level
- Recommended Next Steps
- Questions for a Doctor
- Warning Signs Requiring Immediate Attention



The screenshot displays the 'Results Dashboard' for MediGuide AI, featuring a dark blue sidebar with navigation links: Home, New Assessment, History, Settings, Help, and Disclaimer. The main content area is titled 'Results Dashboard' with the subtitle 'AI-powered guidance based on the information you provided.' It contains seven sections: 1. Symptom Summary (listing symptoms like Headache, Fever, Body Ache, Sore Throat, duration of 2 days, severity level of 7/10, and age/gender of 25/Female); 2. General AI Information (stating that symptoms may be related to common viral infections and are usually self-limiting); 3. Possible Conditions (listing Common Cold, Influenza (Flu), Viral Fever, and Tonsillitis); 4. Urgency Level (indicating a 'Moderate' level based on symptoms); 5. Recommended Next Steps (listing actions like staying hydrated, taking medication, using fever reducers, monitoring symptoms, and consulting a doctor if symptoms persist); 6. Questions for Doctor (listing questions like 'Could this be a viral or bacterial infection?' and 'Do I need any tests?'); and 7. Warning Signs (listing signs like high fever, difficulty breathing, chest pain, severe weakness, persistent vomiting, and symptoms worsening rapidly). A red alarm icon and text at the bottom right of the warning signs section state: 'If any of these occur, seek emergency care immediately.'

MediGuide AI

Home
New Assessment
History
Settings
Help
Disclaimer

MediGuide AI provides general information only and does not replace professional medical advice, diagnosis or treatment advice.

Results Dashboard

AI-powered guidance based on the information you provided.

- 1. Symptom Summary**
Here is a summary of the symptoms and information you provided.
 - Symptoms: Headache, Fever, Body Ache, Sore Throat
 - Duration: 2 days
 - Severity Level: 7 / 10
 - Age: 25 • Gender: Female
- 2. General AI Information**
AI provides general information based on your symptoms.

Your symptoms may be related to common viral infections such as flu or cold. These conditions are usually self-limiting and improve with rest, hydration and basic care. However, this is not a diagnosis.
- 3. Possible Conditions**
Possible conditions related to your symptoms.
 - Common Cold
 - Influenza (Flu)
 - Viral Fever
 - Tonsillitis
- 4. Urgency Level**
Based on your symptoms, the urgency level is:

Moderate

Your symptoms are not severe at the moment but should be monitored. Follow the recommended next steps.
- 5. Recommended Next Steps**
Here are the recommended next steps.
 - ✓ Rest and stay hydrated
 - ✓ Take prescribed medication if any
 - ✓ Use fever reducers if needed
 - ✓ Monitor your symptoms
 - ✓ Consult a doctor if symptoms persist more than 3 days
- 6. Questions for Doctor**
Helpful questions to ask your doctor.
 - Could this be a viral or bacterial infection?
 - Do I need any tests?
 - What medication do you recommend?
 - Are there any complications to watch for?
 - When should I follow up?
- 7. Warning Signs**
Seek immediate medical attention if you experience any of the following.
 - High fever (above 102°F / 39°C)
 - Severe weakness or confusion
 - Difficulty breathing
 - Persistent vomiting
 - Chest pain or pressure
 - Symptoms worsening rapidly

If any of these occur, seek emergency care immediately.

STREAMING & CACHING

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ANALYSIS

Mild Symptoms → LOW Urgency

Fever + Cough → MEDIUM/HIGH Urgency

Severe Symptoms → LOW Urgency

Empty Symptoms → LOW Urgency

Repeated Requests → LOW Urgency

Conclusion

MediGuide AI combines Python, LangChain, OpenAI, and Streamlit to provide structured, safety-focused patient guidance.



THANK YOU

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